IndiaMediaLens

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Chapter 1

Literature Review

Below is a review of ten key papers I found on Aspect-Based Sentiment Analysis (ABSA), focusing on the evolution of ABSA tasks, the development of generative models, and the critical role of dataset annotation.

1.1 Zhang et al. (2021): Aspect Sentiment Quad Prediction as Paraphrase Generation [1]

Dataset Description

This paper introduces the Aspect Sentiment Quad Prediction (ASQP) task. To support this, two new datasets, **Rest15** and **Rest16**, are constructed. These datasets are created by aligning and merging annotations from existing SemEval 2015 and 2016 datasets as well as annotations from other related works.

Annotation

The datasets provide annotations for complete sentiment quadruples: (aspect category, aspect term, opinion term, sentiment polarity). The authors performed significant additional manual annotation to add missing opinion terms for implicit aspects and to resolve conflicts that arose during the merging of different source datasets. In this work, implicit aspect terms (e.g., "it") are explicitly mapped to the pronoun "it", while quadruples with implicit opinion expressions were discarded.

Method Used

The authors propose a novel PARAPHRASE modeling paradigm. This method casts the ASQP task as a sequence-to-sequence (S2S) paraphrase generation problem. A structured sentiment quad is "linearized" into a natural language sentence. For example, the quad ‘(food quality, pasta, over-cooked, negative)‘

is transformed into the target sentence "Food quality is bad because pasta is over-cooked". This target sentence is then generated from the original input sentence using a pre-trained generative model, specifically T5.

Results

The proposed PARAPHRASE method achieved new state-of-the-art results on the ASQP task, significantly outperforming pipeline baselines and another unified generative method, GAS. An ablation study confirmed the importance of generating natural language labels, as model performance dropped when label semantics were removed. The framework also proved effective on the related ASTE and TASD tasks and demonstrated a strong capacity for cross-task knowledge transfer in low-resource settings.

1.2 Cai et al. (2021): ACOS Quadruple Extraction with Implicit Elements [2]

Dataset Description

This work introduces the Aspect-Category-Opinion-Sentiment (ACOS) Quadruple Extraction task. It provides two new datasets: **Restaurant-ACOS** and **Laptop-ACOS**. Restaurant-ACOS is an extension of the SemEval Restaurant dataset. Laptop-ACOS is a new, larger dataset collected from Amazon, noted as being much larger than the corresponding SemEval laptop dataset.

Annotation

The primary contribution of these datasets is the explicit and full annotation of implicit aspects and implicit opinions. In cases of implicit aspects, the aspect term ‘a‘ is set to ‘NULL‘, and the category ‘c‘ defines the target. In cases of implicit opinions, the opinion term ‘o‘ is set to ‘NULL‘, and the sentiment ‘s‘ describes the polarity. The annotation was performed by two PhD students, and a strict quadruple matching F1 score of 75.86% was achieved, indicating substantial agreement.

Method Used

The paper benchmarks the new ACOS task by adapting four representative systems from related tasks. These methods include:

- **Double-Propagation-ACOS:** A rule-based method adapted from Qiu et al. (2011).
- **JET-ACOS:** A pipeline method that first extracts aspect-opinion-sentiment triples using JET (Xu et al., 2020) and then adds a BERT-based category classifier.

- **TAS-BERT-ACOS:** Adapts the TAS-BERT model by changing its tagging schema to predict aspect-opinion pairs (instead of just aspects) conditioned on category and sentiment.
- **Extract-Classify-ACOS:** A two-stage framework that first co-extracts aspect-opinion pairs (including implicit ones) and then uses a classifier to predict the category and sentiment for each pair.

Results

The **Extract-Classify-ACOS** method achieved the highest F1 scores, with 44.61% on Restaurant-ACOS and 35.80% on Laptop-ACOS. The results demonstrated that this method and TAS-BERT-ACOS were capable of handling implicit elements, whereas the other baselines were not. The authors note that the F1 scores are not high, indicating that ACOS (especially with implicit elements) remains a very challenging task.

1.3 Peng et al. (2020): Aspect Sentiment Triplet Extraction [3]

Dataset Description

This paper introduces a new ABSA subtask: Aspect Sentiment Triplet Extraction (ASTE). This task aims to extract a triplet of (Aspect Term, Sentiment Polarity, Opinion Term) to capture "What" is being discussed, "How" the sentiment is, and "Why" (the opinion reason). The datasets used are derived from the SemEval 2014, 2015, and 2016 benchmarks.

Annotation

The authors created new datasets by adding opinion term annotations from Fan et al. (2019) to the existing SemEval data. They merged samples, corrected annotations, and explicitly paired aspects with their corresponding opinion terms to form the ground-truth triplets.

Method Used

A two-stage framework is proposed to solve the ASTE task.

- **Stage 1:** A joint model is used to extract all potential aspect terms (with their sentiments) and all potential opinion terms separately. This stage uses stacked BLSTMs, a GCN, and specialized components like Sentiment Consistency (SC) and Target Guidance (TG) to share information between the two extraction tasks.

- **Stage 2:** All extracted aspects and opinions are fed into a "candidate pair pool". A binary classifier then determines if each aspect-opinion pair is a valid pair. This classifier is enhanced with position embeddings that capture the word-length distance between the aspect and opinion terms.

Results

The proposed two-stage framework established a strong benchmark performance on the new ASTE task. It successfully outperformed several baselines that were adapted from related co-extraction and tagging tasks. Ablation tests confirmed the utility of the model's components, showing that explicitly modeling opinion extraction helped improve aspect extraction performance.

1.4 Scaria et al. (2024): InstructABSA [4]

Dataset Description

This work utilizes several standard ABSA benchmark datasets, including SemEval 2014, 2015, and 2016 (for laptops, restaurants, and hotels) and the ASTE dataset from Peng et al. (2020).

Annotation

The study leverages the existing annotations for a wide variety of ABSA subtasks, including Aspect Term Extraction (ATE), Aspect Term Sentiment Classification (ATSC), Aspect Sentiment Pair Extraction (ASPE), Aspect Opinion Pair Extraction (AOPE), Aspect Opinion Sentiment Triplet Extraction (AOSTE), and Aspect Category Opinion Sentiment Quadruple Extraction (ACOSQE).

Method Used

The paper introduces **InstructABSA**, a paradigm based on instruction learning. The method involves further instruction tuning an existing instruction-tuned model, Tk-Instruct (which is based on T5). The core contribution is the design of the instruction prompts. In addition to the task definition, the prompts are augmented with positive, negative, and neutral examples to guide the model. The main model variant, 'InstructABSA-2', includes two examples for each of the three polarities.

Results

InstructABSA achieved new state-of-the-art (SOTA) results on several key subtasks, surpassing previous SOTA models, including some that were 7 times larger. For instance, it improved the F1 score on the Rest14 ATE subtask by 5.69 points and the accuracy on the Rest15 ATSC subtask by 9.59 points. The method also demonstrated high sample efficiency, achieving competitive results

with only 50% of the training data. An analysis of instruction quality showed that using "delusive" (incorrect) examples caused performance to drop by 10%.

1.5 Röttger et al. (2022): Two Contrasting Data Annotation Paradigms [5]

Dataset Description

This is a conceptual paper that introduces a framework for data annotation. It does not introduce a new benchmark dataset but uses a small, 200-post Twitter dataset for an illustrative experiment. This dataset was sampled from an existing hate speech corpus (Davidson et al., 2017).

Annotation

The paper's central contribution is the definition of two contrasting paradigms for annotating subjective NLP tasks:

- **Descriptive Paradigm:** This approach encourages annotator subjectivity and provides minimal guidelines. The goal is to capture and model the diverse range of valid human beliefs about a subjective task (e.g., "what do different people consider hateful?").
- **Prescriptive Paradigm:** This approach discourages annotator subjectivity by providing highly detailed and specific guidelines. The goal is to encode one specific belief (e.g., a company's content policy) and train models to apply that belief consistently.

The paper argues that dataset creators must explicitly choose one paradigm based on their intended downstream use.

Method Used

The paper's method is an illustrative annotation experiment. 60 annotators were split into three groups to label the same 200 tweets:

- **G1 (Descriptive):** Given a prompt asking for their personal, subjective judgment.
- **G2 (Prescriptive):** Given detailed guidelines and a prompt asking them to apply those criteria objectively.
- **G3 (Control):** Given a simple definition but no detailed guidelines.

Results

The experiment demonstrated a stark contrast in annotator agreement. The descriptive group (G1, $\kappa = 0.20$) and the control group (G3, $\kappa = 0.15$) both showed very low agreement, suggesting they defaulted to their own varied beliefs. The prescriptive group (G2), which used detailed guidelines, achieved substantial agreement ($\kappa = 0.78$). This result validates the hypothesis that comprehensive guidelines are instrumental in discouraging subjectivity and achieving a consistent, prescriptive dataset.

1.6 Chebolu et al. (2023): A Review of Datasets for ABSA [6]

Dataset Description

This paper is a comprehensive survey of datasets for Aspect-Based Sentiment Analysis. It reviews and categorizes 98 publicly available ABSA datasets. These datasets span over 25 domains and include 77 in English and 21 in other languages. The paper provides a centralized GitHub repository summarizing these datasets.

Annotation

The survey organizes the datasets according to the 12 different ABSA subtasks they support (e.g., ATE, ACSA, ASTE, ASQP). It provides a detailed overview of the four main elements annotated in ABSA: aspect terms/targets, aspect categories, opinion phrases, and sentiment polarity. It discusses common annotation procedures, challenges like inconsistent formats (XML, JSON, BIO), and the frequent lack of opinion phrase annotations.

Method Used

The primary methodology is a systematic review of existing ABSA corpora. Additionally, the authors conduct a baseline study to analyze the difficulty and characteristics of different datasets. They test a unified generative paraphrasing model (Zhang et al., 2021a) across six different joint-extraction tasks (ASD, TSD, AOPE, ASTE, TASD, ASQP) on a sample of datasets.

Results

The survey serves as a key resource for researchers, highlighting the scattered nature of ABSA datasets. It identifies several critical gaps and future directions, including: (1) the need for standardization of dataset formats; (2) the need for more datasets in low-resource languages; (3) a call to move from sentence-level to review-level ABSA to better handle context and implicit elements; and (4) the

need to expand beyond product reviews into domains like finance, healthcare, and education.

1.7 Simmering & Huoviala (2023): LLMs for ABSA [7]

Dataset Description

This study uses the gold standard SemEval-2014 Task 4 benchmark dataset, which consists of human-annotated sentences from laptop and restaurant reviews. The authors excluded examples with the "conflict" polarity label to maintain consistency with prior work like InstructABSA.

Annotation

The paper focuses on the joint task of Aspect Term Extraction (ATE) and Aspect Term Sentiment Classification (ATSC). The goal is to extract all aspect terms (e.g., "hard drive") and their corresponding polarities (e.g., "positive").

Method Used

The authors evaluate and compare the performance of OpenAI's GPT-3.5 and GPT-4 models against the previous state-of-the-art, InstructABSA. The LLMs are tested in three settings:

- **Zero-shot:** Using only a task description (system prompt).
- **Few-shot:** Providing 2, 4, 6, or 10 in-context examples in the prompt.
- **Fine-tuned:** Fine-tuning the GPT-3.5 model on the SemEval-2014 training set.

The prompts also utilized OpenAI's function calling feature to enforce a strict JSON output schema.

Results

The fine-tuned GPT-3.5 model achieved a new state-of-the-art F1 score of 83.8% on the joint task, surpassing the previous SOTA (InstructABSA) by 5.7 F1 points. In contrast, the zero-shot and few-shot performance of both GPT-3.5 and GPT-4 was significantly lower (peaking at 65.7% and 71.3%, respectively). The results show that fine-tuning is highly effective and largely eliminates the need for detailed prompts or in-context examples. The paper also provides a cost-performance analysis.

1.8 Yan et al. (2021): A Unified Generative Framework for ABSA [8]

Dataset Description

The framework is evaluated on four widely used ABSA datasets (referred to as $\mathcal{D}_{17}$, $\mathcal{D}_{19}$, $\mathcal{D}_{20a}$, $\mathcal{D}_{20b}$) that are derived from SemEval challenges. These datasets collectively cover seven distinct ABSA subtasks.

Annotation

The paper addresses seven different ABSA subtasks under one framework. These tasks range from single-element extraction, like Aspect Term Extraction (AE) and Opinion Term Extraction (OE), to compound tasks like Aspect-level Sentiment Classification (ALSC), Aspect-Opinion Pair Extraction (Pair), and Triplet Extraction.

Method Used

This paper proposes a unified generative framework that converts all seven ABSA subtasks into a single sequence-to-sequence formulation. The core innovation is to redefine the target output for every subtask as a sequence of indexes.

- **Extraction tasks** (AE, OE) are modeled as generating *pointer indexes* that correspond to the start and end token positions of the aspect/opinion in the input text.
- **Classification tasks** (ALSC) are modeled as generating *class indexes* (e.g., 8 for 'POS', 9 for 'NEG').

Compound tasks like Triplet Extraction generate a mixed sequence of pointer and class indexes. The framework uses a pre-trained BART model as its sequence-to-sequence backbone.

Results

The unified generative framework achieved substantial performance gains, setting new state-of-the-art results on most of the seven subtasks across the four datasets. For example, it significantly outperformed previous models on the complex Triplet Extraction task. The results validate that this unified index-generation formulation is an effective and generalizable end-to-end solution for the diverse set of ABSA subtasks.

1.9 Zhang et al. (2021): Towards Generative ABSA (GAS) [9]

Dataset Description

This work evaluates its proposed framework on four standard benchmark datasets: Laptop14, Rest14, Rest15, and Rest16, which originate from SemEval shared challenges.

Annotation

The framework is designed to be general and is evaluated across four different joint-extraction ABSA tasks:

1. Aspect Opinion Pair Extraction (AOPE)
2. Unified ABSA (UABSA) (aspect term + sentiment)
3. Aspect Sentiment Triplet Extraction (ASTE)
4. Target Aspect Sentiment Detection (TASD) (aspect term, aspect category, sentiment)

Method Used

This paper proposes a unified generative framework (GAS) that tackles various ABSA tasks by formulating them as text generation problems. The authors introduce two distinct paradigms for constructing the target sequence to be generated:

- **Annotation-style:** This paradigm integrates the labels directly into the input sentence. For example, "The pizza was cold" becomes "The [pizza | food quality negative] was cold".
- **Extraction-style:** This paradigm directly generates a linearized string of the desired elements. For example, "(pizza, food quality, negative); (cheese, food quality, negative); ...".

The model uses a pre-trained T5 model. A prediction normalization strategy is also proposed, which uses Levenshtein distance to map slightly incorrect generated words (e.g., "Bbq rib") to the closest valid label ("BBQ rib").

Results

The GAS framework set new state-of-the-art results in almost all cases across all four tasks and datasets. It significantly outperformed previous SOTA models, especially on the complex ASTE and TASD tasks. The extraction-style paradigm was found to be more effective for complex tasks (ASTE, TASD), whereas the annotation-style was slightly better for simpler pair-extraction tasks (AOPE, UABSA).

1.10 Juroš et al. (2024): LLMs for Targeted Sentiment in News Headlines [10]

Dataset Description

This paper focuses on Targeted Sentiment Analysis (TSA) in news headlines. It uses three datasets: STONE (Croatian news headlines), SEN (English and Polish news headlines), and a Spanish dataset of political headlines.

Annotation

The paper explicitly leverages the "descriptive vs. prescriptive" framework from Röttger et al. (2022). The STONE dataset is prescriptive, featuring detailed annotation guidelines and 6 annotator labels per item. The SEN and Spanish datasets are descriptive, with vaguer guidelines that encourage annotator subjectivity.

Method Used

The study compares fine-tuned encoder models (e.g., BERTić for Croatian) against zero- and few-shot LLMs (GPT-4, GPT-3.5, Llama 3, etc.). A primary contribution is the analysis of prompt prescriptiveness. The authors created six levels of prompts for the STONE dataset, ranging from a simple zero-shot instruction (Level 1) to the full, elaborate annotator guidelines with examples (Level 6). The study also evaluates the LLMs' ability to quantify uncertainty (using methods like Self-Consistency Sampling) and compares it to human label variation.

Results

The results showed a clear divide based on annotation style. On the *descriptive* datasets (SEN, Spanish), LLMs (like GPT-4) outperformed the fine-tuned BERT models. However, on the *prescriptive* STONE dataset, the fine-tuned BERTić model was "significantly" better than all LLMs. For LLMs, predictive accuracy generally increased with prompt prescriptiveness from Level 1 to Level 4. However, Levels 5 and 6, which included few-shot examples, sometimes caused a performance drop for larger models like GPT-4, possibly due to sensitivity to example selection.

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